

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
TABLE OF CONTENTS	
INTRODUCTION	iv
SECTION 1 – SYSTEM CONFIGURATION	
1-1 INTRODUCTION	1-3
1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY	1-4
1-2-1 Signa Horizon LX	1-6
1-2-2 Signa Horizon 5.X System	1-9
1-2-3 Signa Advantage 1.5T 5.X & 4.X	1-14
1-3 UPGRADES TO SIGNA HORIZON LX WITH ACGD	1-22
1-3-1 Signa Horizon LX with SGD & Signa MR/i System	1-24
1-3-2 Signa Horizon LX With 8645 Gradients System	1-27
1-3-3 Signa Horizon 5.X System	1-29
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System	1-34
1-3-5 Signa Advantage 4.X System	1-40
1-4 SIGNA HORIZON LX SYSTEM OPTIONS	1-45
1-5 UPGRADES TO SIGNA HORIZON LX WITH SGD SYSTEM	1-48
1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS	1-48
1-7 TRANSPORTABLE/RELOCATABLE UPGRADE	1-49
1-8 TRUE MOBILE UPGRADE	1-49
1-9 VAN MANUFACTURERS CONTACT INFORMATION	1-50
SECTION 2 – ROOM LAYOUTS	
2-1 INTRODUCTION	2-3
2-2 TWO MAGNET SITE LAYOUT	2-4
2-3 MINIMUM DOOR/HALLWAY SIZES	2-4
2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements	2-6
2-4 CABLING CONSIDERATIONS	2-8
2-5 SPECIAL SITING CONSIDERATIONS	2-12
2-5-1 Power Distribution Unit (PDU)	2-12
2-5-2 Blower Box (MG6)	2-13
2-5-3 Water Chiller (WC1) For Gradient Coil Cooling	2-13
2-5-4 SPT Phantom Set Shipping/Storage Cabinet	2-13
2-5-5 Spectroscopy for 1.5T Magnet Only (Optional)	2-14
2-5-6 Magnet Monitor	2-14
2-5-7 System Monitoring & Support Connectivity Requirements	2-15
2-6 ARCHITECTURAL REMINDERS	2-15
2-7 FLOOR LOADING AND WEIGHTS	2-16
2-7-1 Operator Workspace Mounting Requirements For Signa LX	2-17
2-8 COMPONENT DIMENSIONS	2-18
SECTION 3 – MAGNETIC FIELD CONSIDERATIONS	3-1

<u>SECTION</u>	<u>PAGE</u>
SECTION 4 – SITE ENVIRONMENT	
4-1 INTRODUCTION	4-3
4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS	4-3
4-3 COOLING REQUIREMENTS	4-4
4-3-1 Air Cooling	4-4
4-3-2 Water Cooling	4-6
4-4 ALTITUDE	4-6
4-5 LIGHTING	4-6
4-6 ACOUSTICS	4-7
4-7 ROOM VENTILATION	4-11
4-8 CRYOGENIC VENTING	4-11
4-9 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS	4-12
4-10 CONSTRUCTION MATERIALS	4-12
4-11 VIBRATION	4-12
4-11-1 Definition	4-12
4-11-2 Types Of Vibration Image Quality Issues	4-12
4-11-3 Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield	4-13
4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets ..	4-14
4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation	4-14
4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation ..	4-15
SECTION 5 – POWER REQUIREMENTS	
5-1 INTRODUCTION	5-3
5-2 CRITICAL POWER REQUIREMENTS	5-4
5-3 POWER DISTRIBUTION SYSTEM	5-6
5-3-1 Main Disconnect Control (MDC)	5-6
5-3-2 System Power Distribution Unit	5-9
5-4 GROUNDING	5-10
5-4-1 Facility Ground	5-10
5-4-2 System Ground	5-12
5-5 GROUND FAULT PROTECTION	5-14
SECTION 6 – INTERCONNECT DATA	
6-1 INTRODUCTION	6-3
6-1-1 Component Designators	6-4
6-1-2 Definition of Terms	6-5
6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES	6-6
6-2-1 Introduction	6-6
6-2-2 M1033L – Indigo to Octane II with EXCITE Upgrade	6-7
6-2-3 M1060JW – Shield/Cryo Cooler Compressor	6-7
6-2-4 M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure	6-8
6-2-5 M1090JP & M1090JR – Enclosure Upgrade For Fixed Site	6-8
6-2-6 M1090JW – Laser to Halogen Alignment Lights Upgrade	6-9
6-2-7 M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System	6-9
6-2-8 M1090PH – Pre LX to LX Without Phase Array To 4 channel Phase Array Adder Kit For 1.5T System	6-9
6-2-9 M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array	6-10
6-2-10 M1090SJ – 5.5 To LX Upgrade For Fixed Site	6-10
6-2-11 M1191JF – Enclosure Upgrade For Transportable/Relocatable	6-10

SECTION**PAGE****SECTION 6 – INTERCONNECT DATA (Continued)**

6-2-12	M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel	6-10
6-2-13	M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-10
6-2-14	M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet	6-11
6-2-15	M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet	6-13
6-2-16	M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure	6-13
6-2-17	M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System	6-13
6-2-18	M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System	6-13
6-2-19	M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	6-13
6-2-20	M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet	6-14
6-2-21	M1891SP – Collector Kit for Upgrade to 1.0T LCC Magnet	6-15
6-2-22	M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-23	M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet	6-16
6-2-24	M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-25	M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site	6-17
6-2-26	M3090CB – 5.1–5.4 To LX With ACGD Cable Upgrade For Fixed Site ...	6-21
6-2-27	M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site	6-23
6-2-28	M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site	6-24
6-2-29	M3090CH – 3.X–5.4 to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-25
6-2-30	M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel ..	6-25
6-2-32	M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site	6-28
6-2-33	M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-30
6-2-34	M3090EB – 1.5T EXCITE Hardware Kit	6-30
6-3	ADVANTAGE WORKSTATION INTERCONNECTS	6-31
6-4	INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES	6-31
6-5	INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES	6-32
6-6	INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES	6-34
6-6-1	Signa Horizon LX System With SGD & RF/PDU Cabinet	6-34
6-6-2	Signa Horizon LX System With SGD & RF/Penetration Cabinet	6-35
6-6-3	Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet	6-37
6-6-4	Signa Horizon 5.X System	6-39
6-6-5	Signa Horizon 5.X Software & Signa Advantage 5.X System	6-42
6-7	INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE	6-52
6-8	REAR PEDESTAL AND ENCLOSURE CABLE ACCESS	6-54
6-9	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD ..	6-55
6-9-1	M3090DA – 4.X–5.4 MNS Upgrade Kit For ACGD Site	6-55
6-9-2	M3090DB – 5.5 MNS Upgrade Kit For ACGD Site	6-56
6-9-3	M3090DB – LX MNS Upgrade Kit For ACGD Site	6-56
6-10	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD ..	6-57

<u>SECTION</u>	<u>PAGE</u>
SECTION 7 – RF SHIELDED ROOM	
7-1 RF SHIELDED ROOM SPECIFICATION	7-3
7-2 VENTS	7-3
7-3 PLUMBING	7-3
7-4 ELECTRICAL	7-3
7-5 RF PENETRATION PANEL	7-3
7-6 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM	7-5
7-6-1 Magnet Mounting	7-5
7-6-2 Blower Box	7-5
7-7 MAGNET ROOM EQUIPMENT MOUNTING	7-5
7-7-1 Magnet Rundown Unit (MS4)	7-5
7-7-2 Emergency Off Buttons	7-5
7-8 ROOM VENTILATION SWITCH	7-5
SECTION 8 – SHIPPING AND DELIVERY DATA	
8-1 INTRODUCTION	8-3
8-2 UPGRADE CATALOGS SHIPPING INFORMATION	8-6
SECTION 9 – PREINSTALLATION CHECKLIST	
9-1 INTRODUCTION	9-3
9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY	9-3
9-3 GENERAL PREINSTALLATION REMINDERS	9-5
9-4 SAFETY	9-5
9-5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM	9-6
9-6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL	9-7
9-7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET RAMP-UP	9-7
SECTION 10 – TOOLS AND TEST EQUIPMENT	
10-1 INTRODUCTION	10-3
10-5 INSTALLATION EQUIPMENT	10-5
10-7 TEST EQUIPMENT	10-8
10-8 CALIBRATION TOOLS/FIXTURES	10-8
10-9 TOOL KIT	10-9
APPENDIX A – MR SITE VIBRATION TEST GUIDELINES	
A-1 TEST MEASUREMENTS	A-1
A-2 EQUIPMENT (SPECTRAL ANALYZER) SET-UP	A-1
A-3 TEST MEASUREMENTS	A-2
A-3-1 Ambient Baseline Condition	A-2
A-3-2 Normal (Operating) Condition	A-2
A-4 PRESENTATION/INTERPRETATION OF RESULTS	A-3

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a Signa Horizon system upgrade. “Preinstallation work” is done to prepare the customer’s premises for the installation of the upgrade sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as facility input power line to the Main Disconnect Control and Power Distribution Unit. If new wires are installed then the electrical contractor shall ring identify and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non-electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Site renovation.
- Installation of structural reinforcements as required.

All work **MUST** be in compliance with national and local building and safety codes.

All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Control and Power Distribution Unit or the power lines required in the Magnet Room, the Main Disconnect Control option, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, or the RF shielded room specified in this document.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third-party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre-installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of the pre-installation information.